Contents

1 Integral Curves and Flows

1.1 Integral Curves

Definition 1.1.1 ▸ Integral Curve

If V is a vector field on M, an ***integral curve of*** V is a differentiable curve $\gamma : J \to M$ whose velocity at each point is equal to the value of V at that point:

$$\gamma'(t) = V_{\gamma(t)} \quad \text{for all } t \in J.$$

If $0 \in J$, the point $\gamma(0)$ is called the ***starting point of*** γ.

Finding integral curves boils down to sloving a system of ordinary differential equations in a smooth chat. Suppose V is a smooth vector field on M and $\gamma : J \to M$ is a smooth curve. On a smooth coordinate domain $U \subseteq M$, we can write γ in local coordinates as

$$\gamma(t) = \left(\gamma^1(t), \cdots, \gamma^n(t)\right)$$

Then the condition $\gamma'(t) = V_{\gamma(t)}$ for γ to be an integral curve of V can be written

$$\dot{\gamma}^i(t) \left.\frac{\partial}{\partial x^i}\right|_{\gamma(t)} = V^i(\gamma(t)) \left.\frac{\partial}{\partial x^i}\right|_{\gamma(t)},$$

which reduces to the following autonomous system of ordinary differential equatoins:

$$\begin{aligned} \dot{\gamma}^1(t) &= V^1(\gamma^1(t), \ldots, \gamma^n(t)), \\ &\vdots \\ \dot{\gamma}^n(t) &= V^n(\gamma^1(t), \ldots, \gamma^n(t)). \end{aligned}$$

Proposition 1.1.2

Let V be a smooth vector field on a smooth manifold M. For each point $p \in M$, there exists $\varepsilon > 0$ and a smooth curve $\gamma : (-\varepsilon, \varepsilon) \to M$ that is an integral curve of V starting at p.

Proof. The ODE system corosponding to the condition

$$\gamma'(t) = V_{\gamma(t)}$$

is smooth. Thus it satisfies the local Lipschitz condition. Then the proposition follows from the Picard-Lindelöf Theorem. □

1.2 Flows

Definition 1.2.1

We define a ***global flow*** on M (also called a ***one-parameter group action***) to be a continuous left $\mathbb{R}$-action on M; that is, a continuous map $\theta : \mathbb{R} \times M \to M$ satisfying the following properties for all $s, t \in \mathbb{R}$ and $p \in M$:

$$\theta(t, \theta(s, p)) = \theta(t + s, p), \quad \theta(0, p) = p. \tag{1.1}$$

Given a global flow θ on M, we define two collections of maps as follows:

- For each $t \in \mathbb{R}$, define a continuous map $\theta_t : M \to M$ by

  $$\theta_t(p) = \theta(t, p).$$

  The defining properties (1.1) are equivalent to the ***group laws***

  $$\theta_t \circ \theta_s = \theta_{t+s}, \quad \theta_0 = \mathrm{Id}_M. \tag{1.2}$$

  As is the case for any continuous group action, each map $\theta_t : M \to M$ is a homeomorphism, and if the flow is smooth, θ_t is a diffeomorphism.
- For each $p \in M$, define a curve $\theta^{(p)} : \mathbb{R} \to M$ by

  $$\theta^{(p)}(t) = \theta(t, p).$$

The image of this curve is the orbit of p under the group action.

Proposition 1.2.2

If $\theta : \mathbb{R} \times M \to M$ is a smooth global flow, for each $p \in M$ we define a tangent vector $V_p \in T_pM$ by

$$V_p = \theta^{(p)\prime}(0).$$

The assignment $p \mapsto V_p$ is a (rough) vector field on M, which is called the ***infinitesimal generator of*** θ.
The infinitesimal generator V of θ is a smooth vector field on M, and each curve $\theta^{(p)}$ is an integral curve of V.

1.2.1 Fundemental Theorem on Flows

Definition 1.2.3

- If M is a manifold, a ***flow domain*** for M is an open subset $\mathcal{D} \subseteq \mathbb{R} \times M$ with the property that for each $p \in M$, the set $\mathcal{D}^{(p)} = \{t \in \mathbb{R} : (t, p) \in \mathcal{D}\}$ is an open interval containing 0 .
- A ***flow*** on M is a continuous map $\theta : \mathcal{D} \to M$, where $\mathcal{D} \subseteq \mathbb{R} \times M$ is a flow domain, that satisfies the following group laws :
  - For all $p \in M$,
  $$\theta(0, p) = p, \tag{1.3}$$
  - and for all $s \in \mathcal{D}^{(p)}$ and $t \in \mathcal{D}^{(\theta(s,p))}$ such that $s + t \in \mathcal{D}^{(p)}$,
  $$\theta(t, \theta(s, p)) = \theta(t + s, p). \tag{1.4}$$

  We sometimes call θ a ***local flow*** to distinguish it from a global flow as defined earlier. The unwieldy term ***local one-parameter group action*** is also used.

Note.

- ***flows domain*** 要求每一个空间节点上，时间的变化范围都是包含 0 的开区间
- 一个 ***flow*** 就是 M 上活动在 $\mathcal{D}$ 内关于时间变化的一个局部左 $\mathbb{R}$-作用.

Definition 1.2.4

If θ is a flow, we define $\theta_t(p) = \theta^{(p)}(t) = \theta(t, p)$ whenever $(t, p) \in \mathcal{D}$, just as for a global flow. For each $t \in \mathbb{R}$, we also define

$$M_t = \{p \in M : (t, p) \in \mathcal{D}\}, \tag{1.5}$$

so that

$$p \in M_t \Leftrightarrow t \in \mathcal{D}^{(p)} \Leftrightarrow (t, p) \in \mathcal{D}.$$

If θ is smooth, the ***infinitesimal generator of*** θ is defined by $V_p =$

$\theta^{(p)\prime}(0)$.

Note.

- 当我们谈论点在 t 时间后的行为时, 只能对 M_t 上的点讨论.
- 无穷小生成元就是每一点处运动曲线的起始切向量.

Theorem 1.2.5 ▸ Fundamental Theorem on Flows

Let V be a smooth vector field on a smooth manifold M. There is a unique smooth maximal flow $\theta : \mathcal{D} \to M$ whose infinitesimal generator is V. This flow has the following properties:

(a) For each $p \in M$, the curve $\theta^{(p)} : \mathcal{D}^{(p)} \to M$ is the unique maximal integral curve of V starting at p.
(b) If $s \in \mathcal{D}^{(p)}$, then $\mathcal{D}^{(\theta(s,p))}$ is the interval $\mathcal{D}^{(p)} - s = \{t - s : t \in \mathcal{D}^{(p)}\}$.
(c) For each $t \in \mathbb{R}$, the set M_t is open in M, and $\theta_t : M_t \to M_{-t}$ is a diffeomorphism with inverse θ_{-t}.

Note.

- 向量场 V 的最大流 $\theta : \mathcal{D} \subset \mathbb{R} \times M \to M$ 将从 M 中各点出发的极大积分曲线统一表示为 $\gamma_p(t) = \theta(t, p)$. 相对于单纯的一族积分曲线, 最大流 θ 还同时编码了解对初值 p 的光滑依赖, 即 θ 作为 (t, p) 的函数是光滑的
- 直观上, [(c)] 用平移来看. 划定一个区域, 将这些区域整体平移, 删去那些跑到区域外的点, 删去的点正好能补齐因为平移而产生的空缺, 这体现了平移关于时间的可逆性, 平移就是一种特殊的流.

1.2.2 Complete Vector Fields

1.3 Lie Derivatives

Definition 1.3.1

Suppose M is a smooth manifold, V is a smooth vector field on M, and θ is the flow of V. For any smooth vector field W on M, define a rough vector field on M, denoted by $\mathcal{L}_V W$ and called the ***Lie derivative of W***

with respect to V, by

$$(\mathcal{L}_V W)_p = \left.\frac{d}{dt}\right|_{t=0} d(\theta_{-t})_{\theta_t(p)}(W_{\theta_t(p)}) \\ = \lim_{t\to 0} \frac{d(\theta_{-t})_{\theta_t(p)}(W_{\theta_t(p)}) - W_p}{t}, \tag{1.6}$$

provided the derivative exists. For small $t \neq 0$, at least the difference quotient makes sense: θ_t is defined in a neighborhood of p, and θ_{-t} is the inverse of θ_t, so both $d(\theta_{-t})_{\theta_t(p)}(W_{\theta_t(p)})$ and W_p are elements of T_pM
.

Theorem 1.3.2

If M is a smooth manifold and $V, W \in \mathfrak{X}(M)$, then $\mathcal{L}_V W = [V, W]$.

1.4 Commuting Vector Fields

Definition 1.4.1

If θ is a smooth flow, a vector field W is said to be ***invariant under*** θ if W is θ_t-related to itself for each t; more precisely, this means that $W|_{M_t}$ is θ_t-related to $W|_{M_{-t}}$ for each t, or equivalently that $d(\theta_t)_p(W_p) = W_{\theta_t(p)}$ for all (t, p) in the domain of θ.

Theorem 1.4.2

For smooth vector fields V and W on a smooth manifold M, the following are equivalent:
(a) V and W commute.
(b) W is ***invariant under the flow*** of V.
(c) V is ***invariant under the flow*** of W.

Theorem 1.4.3 ▸ Canonical Form for Commuting Vector Fields

Let M be a smooth n-manifold, and let $(V_1, \dots, V_k)$ be a linearly independent k-tuple of smooth commuting vector fields on an open subset $W \subseteq M$.

- For each $p \in W$, there exists a smooth coordinate chart $(U, (s^i))$

centered at p such that $V_i = \partial/\partial s^i$ for $i = 1, \dots, k$.

- If $S \subseteq W$ is an embedded codimension-k submanifold and p is a point of S such that T_pS is complementary to the span of $(V_1|_p, \dots, V_k|_p)$, then the coordinates can also be chosen such that $S \cap U$ is the slice defined by $s^1 = \cdots = s^k = 0$.